CS 201: Data Structures and Algorithms
Course Syllabus, Fall 2026

Unit 3: Graph Traversal

This unit covers the fundamental algorithms for exploring a graph's structure. Breadth-First Search (BFS) explores a graph level by level from a source vertex, using a first-in-first-out (FIFO) queue to track which nodes to visit next. Each node is marked as visited the moment it is enqueued, which prevents the same node from being processed twice through two different paths. Because nodes are dequeued in the exact order of their distance from the source, BFS is the standard technique for finding shortest paths in an unweighted graph.

Depth-First Search (DFS) instead explores as far as possible along each branch before backtracking, typically implemented with a stack (either explicitly or via recursion). DFS is commonly used for tasks such as detecting cycles in a graph and computing a topological ordering of a directed acyclic graph.

Both BFS and DFS run in $O(V + E)$ time on a graph with V vertices and E edges, since each vertex and edge is examined a constant number of times. Students should be able to trace both algorithms by hand on a small graph and implement each one starting from an adjacency list representation.

Prerequisite: Unit 2 (Queues and Stacks). Students should already be comfortable with FIFO queue and LIFO stack operations before starting this unit, since both traversal algorithms depend directly on them.